

wq-sample-label-datasheet-templates

Summary

The purpose of this R markdown is to ...

Some packages that might be of use:

```
library(dplyr)
```

```
##  
## Attaching package: 'dplyr'  
  
## The following objects are masked from 'package:stats':  
##  
##   filter, lag  
  
## The following objects are masked from 'package:base':  
##  
##   intersect, setdiff, setequal, union
```

```
library(stringr)  
library(kableExtra)
```

```
##  
## Attaching package: 'kableExtra'  
  
## The following object is masked from 'package:dplyr':  
##  
##   group_rows
```

The dplyr package is specifically designed to do data manipulation in a way that's relatively easy to read using things called pipes (%>%). See the chapter in this textbook here for some examples of using dplyr.

1. User Inputs

Put info from excel FILLME here as variables. Since we updated the methods we'll probably need to add:

- `random_start` for the starting random number
- `samp_type` (GB,IS, SS, etc.)

Variable names should be lowercase with no spaces (use underscores instead) and not too long (abbreviate where it makes sense).

```
project <- "WWS" #the large project code
study <- "TEST" #the study code
```

2. Bottle Calculator

Based on number of sites and samples per site, calculate bottle types needed as data.frame, then we can print nicely.

3. Label IDs

We'll use this to generate sample IDs and random IDs for labels. For a start just use R functions to take the info from 1 and convert to the IDs. I think you have the most updated things we need but I think we need:

- **SampleID:** project_site_type_date_time
- **RandomID:** project_r###
- **ISCOID:** just the number order it goes in the isco?

```
#paste function works well to join things together
paste0("A", "B", "C")
```

```
## [1] "ABC"
```

```
paste("A", "B", "C")
```

```
## [1] "A B C"
```

```
paste("A", "B", "C", sep="_")
```

```
## [1] "A_B_C"
```

```
#the seq function is good for generating a sequence
seq(1,4)
```

```
## [1] 1 2 3 4
```

```
seq(as.Date("2023-02-01"), as.Date("2024-01-03"), by="month")
```

```
## [1] "2023-02-01" "2023-03-01" "2023-04-01" "2023-05-01" "2023-06-01"
## [6] "2023-07-01" "2023-08-01" "2023-09-01" "2023-10-01" "2023-11-01"
## [11] "2023-12-01" "2024-01-01"
```

```
#we can store all this in a data.frame similar to excel
data <- data.frame(project = project,
                   study=study,
                   dates = seq(as.Date("2023-02-01"),
                              as.Date("2024-01-03"), by="month"))

data
```

```
##      project study      dates
## 1      WWS  TEST 2023-02-01
## 2      WWS  TEST 2023-03-01
## 3      WWS  TEST 2023-04-01
## 4      WWS  TEST 2023-05-01
## 5      WWS  TEST 2023-06-01
## 6      WWS  TEST 2023-07-01
## 7      WWS  TEST 2023-08-01
## 8      WWS  TEST 2023-09-01
## 9      WWS  TEST 2023-10-01
## 10     WWS  TEST 2023-11-01
## 11     WWS  TEST 2023-12-01
## 12     WWS  TEST 2024-01-01
```

4. Specifying analyte

We might need to do some experimenting here. For a set of ISCO samples or a small number of grab samples I think we could just let the user specify the analytes they plan to run (i.e., `analyte <- c("DOC", "DOM")`). If we want to get fancier and let people run only a subset of samples for something we'll need to get fancier. But for now maybe we just allow all or nothing.

5. Generating the .csv for labels

We'll want to write the labels we need to a .csv file so we can use it for making labels. We'll want to just pull the columns we want.

```
print_data <- data %>% select(project, study)
print_data
```

```
##      project study
## 1      WWS  TEST
## 2      WWS  TEST
## 3      WWS  TEST
## 4      WWS  TEST
## 5      WWS  TEST
## 6      WWS  TEST
## 7      WWS  TEST
## 8      WWS  TEST
## 9      WWS  TEST
## 10     WWS  TEST
## 11     WWS  TEST
## 12     WWS  TEST
```

```
#write.csv(print_data, "filename.csv", row.names=FALSE)
```

6. Making the tables

This part will be a little more complicated, since we want specific formatting. I think if we use the LaTeX command below (which is what the Rmarkdown uses to convert to PDF), it should start a new page that

we can format how we like. I'm new to this part too, so we could do some research together when you get here.

`\newpage`

Date: _____

Technician: _____

Project Code: _____

Study Code: _____

project	study	dates
WWS	TEST	2023-02-01
WWS	TEST	2023-03-01
WWS	TEST	2023-04-01
WWS	TEST	2023-05-01
WWS	TEST	2023-06-01
WWS	TEST	2023-07-01
WWS	TEST	2023-08-01
WWS	TEST	2023-09-01
WWS	TEST	2023-10-01
WWS	TEST	2023-11-01
WWS	TEST	2023-12-01
WWS	TEST	2024-01-01